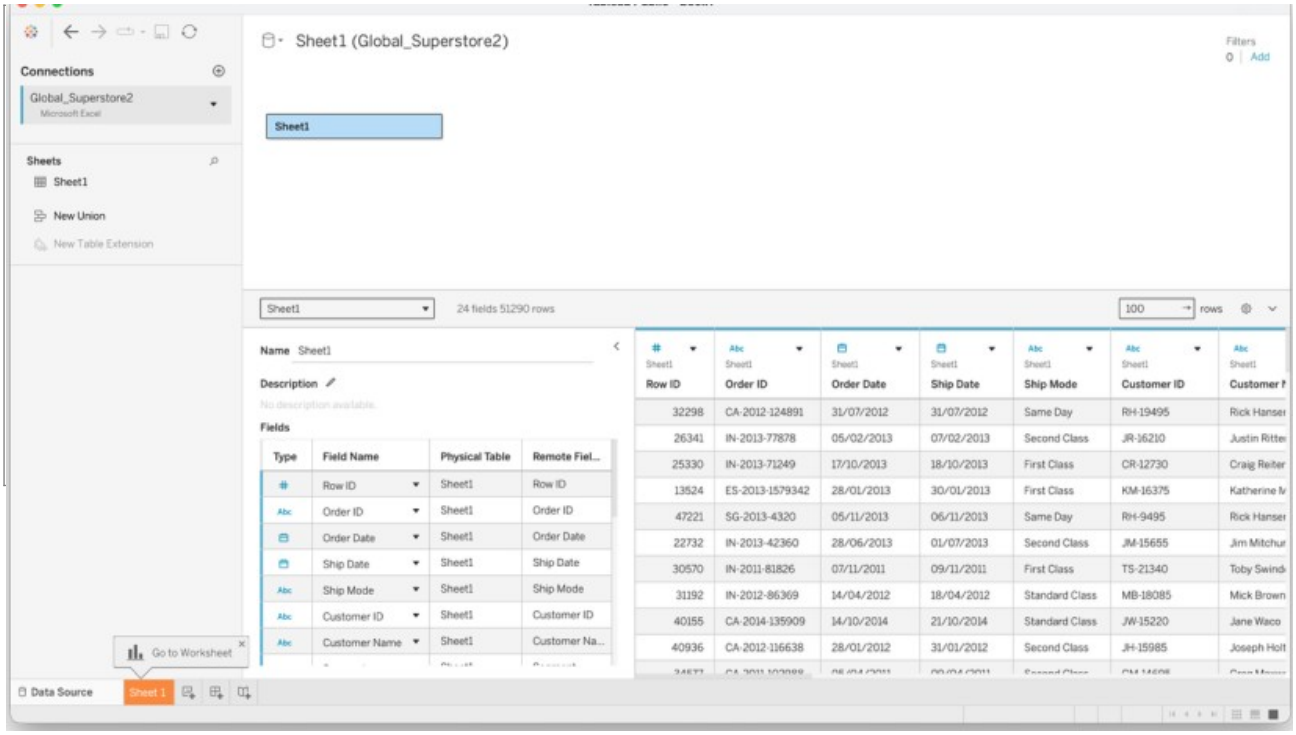
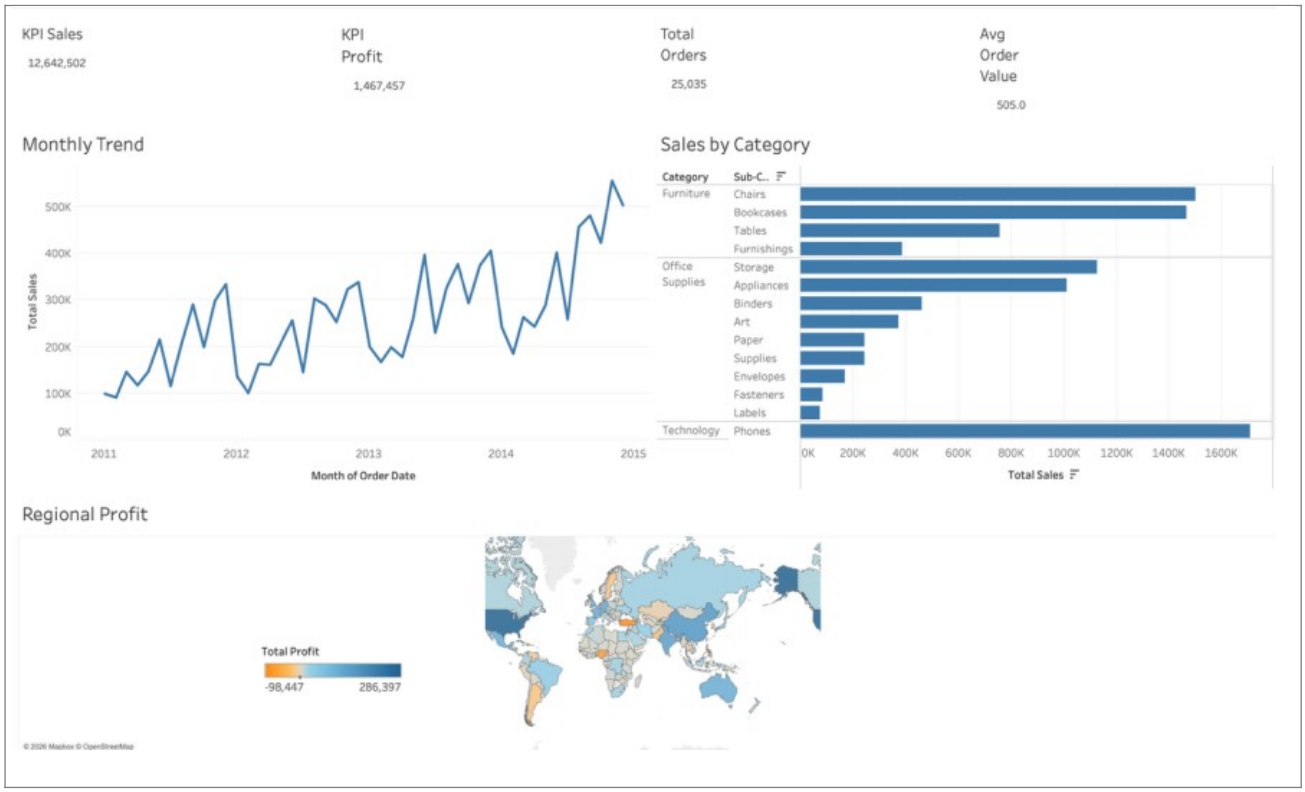
Lab Sheet – 05

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Dataset in Tableau:



Final Dashboard:



Questions and Answers

1. What is Business Intelligence (BI)? How does it support organizational decision-making?

Business Intelligence is the set of tools, methods and processes a company uses to collect data from different places, clean it, store it and turn it into easy reports, charts and dashboards. In simple words, it changes boring raw data into useful information that people can understand quickly.

How it helps in decision-making:

- **Facts instead of guesses:** managers can check real numbers before taking a decision, like which product to promote or which branch needs help.
- **Past and present view:** it shows what happened earlier (last year's sales) and what is happening now, so trends are easy to see.
- **Early warning:** problems like falling sales or rising costs are noticed early, before they become big.
- **Better planning:** past patterns help in forecasting demand, setting targets and planning budgets.
- **One version of truth:** everyone in the company looks at the same data, so there are fewer arguments and faster decisions.

2. What is an interactive dashboard? How is it different from a static report?

Interactive dashboard: a single screen that shows important numbers (KPIs) and charts together, and the user can play with it. Clicking on a bar, choosing a year from a dropdown, or hovering over a point changes or explains the other charts instantly. We can also go deeper into the data, for example from a country to its states to its cities.

Static report: a fixed document like a PDF, printed sheet or a slide, made at one point of time. It looks the same to everybody, and if someone wants to see something extra, a new report has to be prepared.

Point	Interactive dashboard	Static report
User control	User can filter, click and drill down	No control, only reading
Data	Can be refreshed and updated	Fixed at the time it was made
Questions	New questions can be answered on the spot	Needs a new report each time
Example	Tableau or Power BI dashboard	PDF monthly sales report

3. Compare Tableau and Microsoft Power BI based on features, usability, and business applications.

Basis	Microsoft Power BI	Tableau
Features	Works smoothly with Excel, Microsoft 365 and Azure. Uses Power Query to clean data and DAX for calculations.	Very strong in visualization with a wide range of charts and maps. Handles big and varied data well. Uses calculated fields and its own VizQL engine.
Usability	Looks like Microsoft Office, so it is easy for people who already use Excel. Good for beginners.	Drag and drop on shelves gives a lot of freedom to design, but it takes a bit more time to master.

Basis	Microsoft Power BI	Tableau
Cost	Generally cheaper, so popular for company-wide use.	Generally costlier, but Tableau Public is free for learning (with public sharing).
Business use	Best for companies already using Microsoft products, and for regular reporting for many employees.	Best for analysts who need deep data exploration, creative visuals and storytelling.

4. What are Key Performance Indicators (KPIs)? Give examples of KPIs used in sales analytics.

KPIs are measurable values that show how well a company, team or process is doing in reaching its goals. They work like a report card: one look tells whether we are on track or not.

Examples of KPIs in sales analytics:

- **Total Sales (Revenue):** the total money earned from sales in a period.
- **Total Profit and Profit Margin:** money left after costs, and that profit as a percentage of sales.
- **Number of Orders:** how many orders were placed.
- **Average Order Value:** total sales divided by number of orders, which shows how much a customer spends on average per order.
- **Sales Growth Rate:** how much sales increased or decreased compared with the previous period.
- **Customer Acquisition Cost and Customer Lifetime Value:** what it costs to get a new customer and how much that customer is expected to bring over time.

5. Explain the purpose of filters, slicers, and drill-down functionality in dashboards.

- **Filters:** they limit the data shown to only what we need, such as only the year 2014 or only the Furniture category. Filters can be set on one chart, a page or the full dashboard.
- **Slicers:** these are visible controls on the dashboard, like dropdowns, buttons or date sliders, that let any user change what the dashboard shows with one click, without touching the settings.
- **Drill-down:** it lets us move from a summary to more detail step by step. For example, Year to Quarter to Month, or Category to Sub-Category to Product. We first see the big picture and then find out exactly where a change came from.

6. What factors should be considered while designing an effective business dashboard?

- **Know the audience:** a top manager needs a quick summary of main KPIs, whereas an analyst may need detailed data to explore.
- **Clear layout:** keep the most important numbers (KPI cards) at the top, then trends, and then details like maps or tables.
- **Right chart for the data:** line charts for trends over time, bar charts for comparisons, maps for locations.
- **Smart use of colour:** use a few consistent colours with a meaning, and avoid red-green combinations that colour-blind people cannot tell apart.

- **Keep it simple:** too many charts and decorations confuse the reader. Every visual should have a purpose.
- **Interactivity and speed:** add useful filters and make sure the dashboard loads fast.

7. How can dashboards help managers monitor organizational performance in real time?

- **Live data connection:** when a dashboard is connected directly to a live database (for example Tableau live connection or Power BI DirectQuery), the numbers update automatically as new sales or transactions come in.
- **Alerts:** a manager can set limits, like "tell me if daily sales fall below a set value", and the system sends a notification when it happens.
- **Quick action:** since the manager sees problems as they happen, they can react the same day instead of waiting for a monthly report.

8. Why is data visualization important in Business Intelligence tools?

Raw data in big tables is very hard to read. Visualization turns it into charts, maps and graphs which our brain understands much faster. With a good chart we can easily spot trends, seasonal ups and downs, unusual values (outliers) and relationships between things. It also helps people who are not technical to understand the data and to take decisions confidently. For example, a line chart of monthly sales shows the festival-season rise immediately, while finding it in thousands of table rows would take a long time.

9. What insights can be derived from a sales dashboard for an e-commerce company?

- **Seasonal patterns:** which months or festival periods have the highest sales, so stock and staff can be planned in advance.
- **Best and weakest products:** which categories and sub-categories earn the most, and which ones sell a lot but give little or negative profit.
- **Regional performance:** which countries or regions give the most sales and profit, and where marketing money should be increased.
- **Discounts and returns:** whether heavy discounts or product returns are reducing profit.
- **Customer behaviour:** average order value, repeat customers, and preferred shipping modes.

10. Explain how interactive dashboards improve communication between data analysts and business stakeholders.

Earlier, analysts used to send reports and stakeholders had to wait for another report whenever they had a new question. An interactive dashboard changes this into a shared tool that both sides can use together.

- Stakeholders can explore the data themselves by using filters and drill-downs, so they get answers faster.
- In a meeting, the analyst can change a filter or parameter live and show "what if" results right away.
- Everyone looks at the same visuals and numbers, so there is less confusion and fewer misunderstandings.
- Analysts save time on repeated report requests and can focus on deeper analysis.